Project Synopsis: Deep Learning & XAI for Multi-Class Alzheimer’s Disease Detection

1. Objective

The primary goal of this project was to develop a highly accurate, clinically reliable, and interpretable machine learning pipeline capable of classifying Alzheimer's disease into four distinct progression stages using medical imaging (MRI scans). The project bridges the gap between high-performance computer vision and medical transparency.

2. Dataset & Preprocessing

The model analyzes brain scans categorized into four classes: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented.

- **Data Augmentation:** Implemented custom MRI augmentation layers to prevent overfitting and ensure the model generalizes well to real-world anatomical variations.
- **Pipeline:** Images were resized to 224x224 and passed through dedicated preprocessing functions specifically calibrated for the chosen base architecture.

3. Model Architecture

The core classification engine utilizes **Transfer Learning** via a deep Convolutional Neural Network (CNN).

- **Base Model:** Utilized the highly proven **ResNet50** architecture, leveraging its pre-trained weights for advanced feature extraction (identifying textures, shapes, and contrasts in the brain tissue).
- **Fine-Tuning:** The base ResNet layers were frozen, and a custom, trainable classification head was built on top to map the extracted visual features to the four specific dementia stages.

4. Performance & Statistical Evaluation

The model achieved exceptional clinical-grade metrics, rigorously validated against standard and advanced statistical measures:

- **Global Accuracy:** ~98% across all validation samples.
- **F1-Scores:** Reached 1.00 for ModerateDemented and 0.99 for MildDemented. The model successfully navigated the most difficult clinical boundary (NonDemented vs. VeryMildDemented) with 0.96+ F1-Scores.
- **Advanced Metrics:** Proved extreme robustness with a **Matthews Correlation Coefficient (MCC) of 0.9697** and a **Cohen's Kappa of 0.9695**, proving the model is genuinely learning underlying disease features rather than relying on class imbalances or random chance.
- **Confidence Calibration:** Successfully aligned the model's predictive probability (Average Confidence) with its actual accuracy, ensuring the AI is not dangerously overconfident on ambiguous scans.

5. Explainable AI (XAI) Integration

To transition the model from a "black box" to a clinically trustworthy tool, **Grad-CAM (Gradient-weighted Class Activation Mapping)** was implemented.

- The XAI pipeline extracts gradients from the final convolutional layer of the ResNet50 base.
- It generates color-coded heatmaps superimposed over the original MRI scans (masking out the black background for clarity).
- This provides visual proof that the model is diagnosing based on legitimate anatomical regions (like the ventricles or hippocampus) rather than image artifacts.

6. Deployment & User Interface

The entire pipeline was successfully wrapped into an interactive, real-time web application using **Gradio**.

- The interface allows a user to upload a raw MRI scan.
- In real-time, it outputs the exact probability scores across all four dementia stages.
- Simultaneously, it generates and displays the XAI Grad-CAM heatmap, providing an immediate, explainable second opinion for medical professionals.